
Phase 1: Initial Plan and Requirement Gathering Document for Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

Version 1.0

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COSC 336: Introduction to Software Engineering

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Revision history

Name	Date	Reason For Changes	Version

Adapted from the supplied SRS template for Phase 1. Sections 4 and 5 retain its functional and nonfunctional requirements organization. Later phases will iterate on this initial baseline version.

1. Introduction and Project Overview

1.1 Project Overview and Objectives

The project brief describes universities purchasing resources while similar usable items remain idle elsewhere. Existing asset records and management systems may identify ownership and location without providing a convenient way to discover suitable resources, arrange their reuse, and follow their condition throughout their life cycle. Finding an item is therefore only part of the problem: departments also need reliable information and a clear process for obtaining it.

The proposed system is an internal university platform that connects departmental resource needs with available campus assets. It will support registration, search, requests, reservations, approvals, transfers, maintenance, and life-cycle records. AI-assisted classification, semantic matching, sustainable-action recommendations, a conversational assistant, and generated reports will extend these conventional functions.

The project aims to make suitable reuse easier for requesting departments and surplus-resource handovers more manageable for supplying departments. It also aims to improve the reliability and traceability of asset information, support appropriate repair and end-of-life decisions, and estimate financial and environmental benefits using stated assumptions. Conventional and AI-enhanced approaches will be compared to establish where AI adds value and where explicit rules and human judgment remain necessary.

1.2 Document Purpose

This Phase 1 document records Group 6's initial understanding of the project problem, scope, stakeholders, and requirements. It sets out the proposed success criteria and preliminary delivery plan for a platform serving universities. The document forms a starting point for feasibility assessment and subsequent requirements development; assumptions and unresolved details will be reviewed as the project progresses.

2. Project Scope

The project covers an internal resource-exchange and asset-management platform for universities. It will support campus resources such as furniture, computers, electronic devices, laboratory equipment, books, and office materials. Its scope extends from asset registration and publication through requests, reservations, approved transfers, inspection, maintenance, repair, refurbishment, and recorded end-of-life actions.

The system will maintain relevant ownership, custody, location, condition, and history information. Conventional and AI-assisted functions will support resource discovery, matching, decisions, and reporting. Access will depend on assigned roles, and AI recommendations will not authorize actions or bypass required approvals.

University personnel will perform physical activities such as collection, storage, inspection, and repair. The application will coordinate these activities and record their outcomes. Relevant costs and transfer arrangements are included, but complete procurement, accounting, transport, and

warehouse-management systems are outside the project's scope.

For initial planning, the prototype is assumed to maintain its own records without requiring live integration with existing university systems. Representative sample data may be used for demonstration. Development is constrained by the four-person team and the semester schedule; available resources, service costs, data access, and potential integrations will be assessed during feasibility analysis. These assumptions will be revised as the project develops.

3. Stakeholder Analysis and Requirements Gathering

3.1 Stakeholders, Needs, and Responsibilities

The project brief identifies the principal operational roles below. Their proposed responsibilities support resource exchange and asset management in universities, detailed delegations will be refined during requirements analysis.

Department representatives and asset custodians need a manageable way to publish surplus resources and maintain accurate information. They register and update assets within their responsibility, including condition, availability, location, and supporting records. Authorized representatives approve outgoing reservations and transfers and confirm receipt on behalf of the receiving department.

Requesters need suitable resources and clear progress information. They view published asset details, submit departmental needs, compare conventional search results and AI recommendations, request available resources, and track their requests. Requester access alone does not permit release approval.

Business administrators, procurement officers, and finance officers support institutional oversight. Business administrators review exceptional allocation issues and authorize donation, recycling, or disposal within their delegated authority. Procurement officers use outstanding requests and reuse information to inform purchasing decisions. Finance officers review asset values, relevant costs, and estimated savings. Access to operational and financial records depends on the assigned role.

Maintenance and facilities personnel need asset identifiers, locations, condition information, and assigned work details. Maintenance staff record inspections, defects, repairs, refurbishment, costs, and resulting condition changes. Facilities personnel coordinate physical collection and storage where required, with arrangements and handovers recorded in the system.

Sustainability officers need reliable activity records and transparent calculation assumptions. They review reuse, repair, waste-diversion, and estimated emissions indicators, assess recommended actions, and prepare sustainability summaries.

System administrators manage accounts, departmental membership, permissions, and technical operation. They need administrative records and relevant logs. Technical access does not automatically grant authority to approve business transactions.

Group 6 and the academic reviewers are project stakeholders. The team needs clear requirements, manageable responsibilities, and traceable decisions to develop and evaluate the prototype. The professor and lab instructors need documentation and demonstrations that show compliance with the assignment, sound engineering reasoning, and individual contributions.

3.2 Current Process and Proposed Workflow

For initial planning, the following baseline represents the process to be improved. It is informed by the project brief and current systems at Khalifa University (KU), and will be validated as requirements develop.

Departments identify surplus resources through manual inventory reviews, physical checks, or staff reports of unused equipment. A department needing a resource submits a request to the responsible staff, who check available records and coordinate with other departments or procurement. Transfers require authorization, collection arrangements, and updates to ownership, custody, and location records. Maintenance staff record inspections, repairs, costs, and condition changes. Donation, recycling, and disposal follow the relevant approval process, after which asset records are updated.

These activities may involve separate registers, forms, and communications on different platforms. Consequently, a recorded asset may not be easily discoverable by another department, and its availability or condition may require additional confirmation. Approval, collection, and maintenance information may also be difficult to follow across the complete life cycle.

The proposed system will connect these activities within a shared workflow. Authorized staff publish available resources, departments submit needs, and users review search results or AI-recommended matches. Approved reservations and transfers are tracked through receipt, with the relevant asset records updated. Inspection, repair, refurbishment, and end-of-life actions remain linked to the asset's history. AI supports classification, matching, and recommendations while authorized personnel retain responsibility for approvals and physical actions.

Resource exchange: from availability to accepted receipt

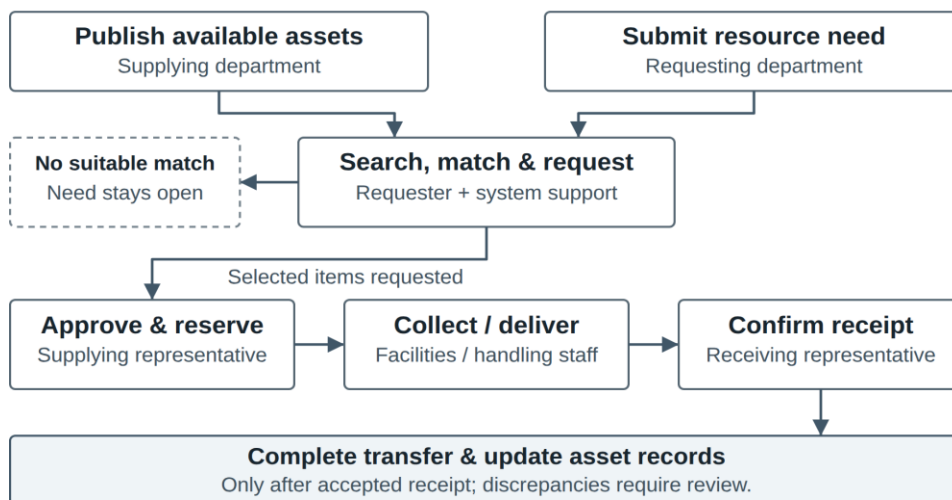


Figure 1. Proposed resource-exchange workflow.

3.3 Gathering Methods and Findings

Initial requirements were gathered through analysis of the project brief, Phase 1 slides, and supplied SRS template. Current processes were analyzed using internal policy documents of KU. The brief established the required capabilities and project boundaries, while the slides and template guided the organization of the initial specification. Lecture material supported the distinction between functional requirements, quality expectations, and later design decisions.

AI-assisted scenario walkthroughs examined resource publication, requests, matching, approvals, collection, receipt, and maintenance. These discussions considered both requesting and supplying departments, including incomplete asset information, competing requests, and unsuitable recommendations.

The analysis identified the need for manageable data entry, clear responsibility for updating records, controlled reservations and transfers, and traceable life-cycle information. It also established that discovering an asset does not confirm its availability or suitability, and that an AI recommendation does not authorize an action. These findings inform the initial requirements and proposed operating rules.

The internal KUBee chatbot was also consulted for background information about asset management processes in KU. No formal stakeholder interviews or surveys have been conducted; operational assumptions remain subject to further validation.

3.4 Requirements Prioritization

Requirements are initially prioritized according to their importance to the project objectives, dependencies, and the consequences of failure. Priorities guide development order and effort; they do not remove functionality required by the project brief.

High-priority requirements establish a dependable exchange workflow: user access control, asset registration and publication, resource requests, search, reservations, approvals, transfers, and life-cycle and maintenance records. AI-assisted classification, semantic matching, ranking, and explanations are also high priority because they demonstrate the project's central comparison between conventional and AI-enhanced approaches. Data validation and authorization apply throughout these functions.

Ease of use for both requesting and supplying departments is a high priority across the system. Publication, requests, and handovers should minimize repeated data entry and unnecessary steps. Workflows should make responsibilities, approval procedures, current status, and the next required action clear, so users can participate without having to work out the process themselves.

Medium-priority requirements build on these foundations, including sustainable-action recommendations, the conversational assistant, sustainability indicators, notifications, and operational and AI-generated reports. These remain required capabilities, but their integration depends on reliable records and functioning workflows. Their technical feasibility will be investigated early rather than postponed until implementation.

Low priority is reserved for optional refinements beyond the required functionality. Such refinements will be considered only after the required capabilities are covered.

These priorities will be reviewed during feasibility and detailed requirements analysis as dependencies, available resources, and technical risks become clearer.

4. System Features

4.1 User, Department, and Access Management

Priority: High

This feature establishes departmental membership and controlled access. A system administrator manages departments and user accounts, assigns roles, and maintains permissions. Users sign in to access the functions and records permitted by their roles; invalid credentials or unauthorized actions are rejected.

1. *The system shall allow authorized system administrators to create and update department records.*
2. *The system shall allow authorized system administrators to create, update, and deactivate user accounts and assign departmental membership and roles.*
3. *The system shall authenticate users before granting access to protected functions and allow authenticated users to sign out. Invalid credentials shall not establish an authenticated session.*
4. *The system shall restrict access to records and actions according to the user's assigned permissions, including interactions through the AI assistant. Technical administration privileges shall not automatically grant business approval authority.*
5. *The system shall record changes to accounts, departmental membership, and assigned permissions, identifying the responsible user, action, and time.*

4.2 Asset Registration and Publication

Priority: High

This feature allows departments to record assets and make surplus resources discoverable. An authorized representative enters or updates an asset record and publishes it when the required information is present. The system reuses existing details, identifies missing information, and displays the listing's condition and availability.

6. *The system shall assign each asset record a unique identifier and support its name, description, category, quantity, owner, department, custodian, location, purchase date, estimated value, condition, availability, photographs, and supporting documents.*
7. *The system shall allow authorized representatives to publish and update surplus, underutilized, or unwanted assets within their responsibility without central pre-publication approval. A listing may be withdrawn only when no active reservation or transfer applies.*
8. *The system shall reuse saved asset information when preparing a listing, allowing authorized users to correct or complete it without entering the same details again.*
9. *The system shall prevent publication when the asset name, positive quantity, owner, department, custodian, location, or declared availability is missing or invalid. It shall identify the problem while preserving the information already entered.*
10. *The system shall display recorded condition and availability and clearly identify unknown condition or incomplete information. Publication alone shall not indicate that an asset is approved for transfer.*

11. *The system shall retain a record of changes to asset details, identifying the responsible user, time, and information changed.*

4.3 Resource Requests, Search, and Reservations

Priority: High

A requester describes a departmental need, searches available resources, and expresses interest in suitable items. The system carries existing request information forward, tracks progress, and confirms a reservation only after authorized approval and an availability check.

12. *The system shall allow requesters to submit departmental resource requests specifying category, purpose, specifications, quantity, preferred condition, urgency, location, and required date. It shall identify missing required information, non-positive quantities, or invalid dates while preserving entered information.*
13. *The system shall provide keyword search and filtering by category, condition, department, location, quantity, availability, asset age, estimated value, and life-cycle status. When a search returns no results, it shall indicate this without closing the resource request.*
14. *The system shall allow requesters to express interest in a listed asset and link it to an existing departmental request without re-entering that request's details. An expression of interest shall not reserve the asset.*
15. *The system shall confirm a reservation only after approval by an authorized representative and a check of current availability. Confirmed reservations shall not exceed the available quantity, including when requests compete for the same resources.*
16. *The system shall display request and reservation status, allocated and outstanding quantities, and the next required action and responsible role. If a requester accepts partial fulfilment, the remaining need shall stay open unless explicitly closed.*
17. *The system shall allow requesters to cancel requests or reservations before dispatch and release any associated reserved quantity. Once dispatch is recorded, cancellation shall not return the affected resources to available stock until custody and receipt are reconciled.*

4.4 Approval and Transfer Management

Priority: High

Authorized representatives review reservation, release, and transfer requests. Related approvals may be completed together where the representative has the required authority, while each decision remains recorded. The system tracks collection or delivery through receipt confirmation and updates the asset records accordingly.

18. *The system shall record approval or rejection of reservation, release, and transfer requests, identifying the authorized decision-maker, date, and reason.*
19. *The system shall maintain a transfer record linking the assets, approved quantities, supplying and receiving departments, collection or delivery arrangements, responsible personnel, destination, and supporting documents.*
20. *The system shall prevent release for ordinary use when required approvals are missing, mandatory suitability information remains unresolved, or the asset is recorded as unsafe or unusable. It shall identify the issue requiring resolution.*
21. *The system shall allow an authorized receiving representative to record the quantity received, observed condition, and acceptance or discrepancy. Discrepancies shall be*

flagged for resolution, with actual custody and location recorded while the transfer remains unresolved.

22. *The system shall mark a transfer complete only after accepted receipt of the approved quantity. It shall update ownership where authorized, custody, location, and availability, and retain the transfer history. Scheduling collection or recording dispatch alone shall not complete the transfer.*

4.5 Asset Life-Cycle and Maintenance Tracking

Priority: High

This feature maintains an asset's history beyond its initial listing or transfer. Authorized users report defects and request service; maintenance personnel record findings, actions, costs, and condition changes. End-of-life actions require authorization and remain part of the asset record.

23. *The system shall maintain a chronological history of acquisition, allocation, utilization, inspection, maintenance, repair, transfer, refurbishment, donation, component recovery, recycling, retirement, and disposal events, identifying the date and responsible user.*
24. *The system shall allow authorized users to report defects, request inspection or repair, and track the status of those requests against the relevant asset.*
25. *The system shall allow maintenance personnel to record inspection findings, maintenance, repair or refurbishment actions, associated costs, outcomes, and the asset's condition after service.*
26. *The system shall keep assets recorded as unfit for ordinary use unavailable for ordinary-use reservations until an authorized condition update establishes their suitability. Returning an asset to service shall not override an existing reservation or transfer.*
27. *The system shall require authorized approval before recording donation, component recovery, recycling, or disposal as completed. It shall retain the decision, reason, date, responsible personnel, and supporting records.*
28. *The system shall prevent retired or disposed assets from appearing as available for reservation while retaining their historical records for authorized review and reporting.*

4.6 AI-Assisted Asset Classification

Priority: High

This feature reduces manual classification and improves consistency in asset records. When a user supplies asset information, the system suggests relevant metadata for review. Suggestions remain distinguishable from confirmed information, and users can continue manually if assistance is unavailable.

29. *The system shall analyze supplied asset titles, descriptions, specifications, images, and documents to recommend categories, subcategories, standardized tags and descriptions, and possible material types, and identify missing metadata.*
30. *The system shall allow authorized users to accept, correct, or reject suggested metadata within the registration or editing workflow. AI suggestions shall not overwrite confirmed asset information without user acceptance.*
31. *The system shall identify uncertain or unsupported suggestions and retain the service or model version used and the user's acceptance or correction outcome.*
32. *If classification assistance fails or cannot produce a usable suggestion, the system shall explain that assistance is unavailable and allow manual entry without losing the information already supplied.*

4.7 Semantic Matching and Recommendation Ranking

Priority: High

This feature helps departments find suitable resources even when listings and requests use different terminology. The system compares available assets with a request, ranks potential matches, and explains its recommendations. Users review the results before proceeding through the normal request and approval workflow.

33. *The system shall match departmental requests with available asset listings using semantic meaning rather than relying only on identical keywords.*
34. *Matching shall consider category, functional purpose, technical specifications, quantity, asset condition, requested condition, location, urgency, required date, transfer cost, and repair cost.*
35. *The system shall rank suitable matches according to functional suitability, urgency, transfer cost, repair requirements, waste-diversion benefit, and estimated carbon-emission reduction. Confirmed incompatibilities with mandatory requirements shall not be presented as fully suitable matches.*
36. *Each recommendation shall display a compatibility score and a readable explanation of the main influencing factors, including uncertainty or missing information. Missing mandatory information shall not be treated as a confirmed match.*
37. *The system shall allow users to accept or reject recommendations and retain the recommendation, service or model version, and review outcome. Selecting a recommendation shall not bypass reservation, availability, or approval rules.*
38. *If no suitable match is found or the matching service is unavailable, the system shall explain the outcome and allow conventional search without automatically closing the departmental request.*

4.8 Sustainable-Action Recommendations

Priority: Medium

This feature provides AI-assisted advice on an asset's next appropriate action. It uses available records and estimates to explain the options, while authorized personnel remain responsible for deciding and approving what happens.

39. *The system shall recommend retention, transfer, repair, refurbishment, donation, component recovery, recycling, or retirement, with disposal considered as a final option.*
40. *Recommendations shall consider recorded condition, maintenance history, departmental demand, relevant transfer and repair costs, and potential financial and environmental benefits.*
41. *The system shall explain the recommended action and identify the supporting information, assumptions, and uncertainties. When essential information is missing, it shall indicate the need for further review rather than present a definitive recommendation.*
42. *The system shall allow authorized users to accept, reject, or override recommendations and record their decision, reason, and the service or model version used. Accepting a recommendation shall not execute the action or bypass its required approvals.*

4.9 Conversational Assistant

Priority: Medium

An LLM-powered assistant will help users interact with the system in natural language. It will support common tasks using accessible records and supplied guidance, while keeping requests and actions within the normal workflow.

43. *The assistant shall support natural-language resource searches, departmental request preparation, questions about asset history and request status, policy guidance, sustainability information, and navigation to relevant system functions.*
44. *When preparing a request, the assistant shall allow the user to review and correct the interpreted details before submission through the standard request workflow.*
45. *The assistant shall base factual answers on records accessible to the user and policy information supplied to the system. It shall indicate when information is unavailable or insufficient rather than present an unsupported answer as confirmed.*
46. *If the assistant is unavailable or cannot complete a task, the system shall allow the user to continue through conventional search, forms, and status pages.*

4.10 Reporting and Sustainability Measurement

Priority: Medium

Authorized users will review operational activity and estimated financial and environmental outcomes. Conventional reports will present recorded data and calculated indicators, while AI-generated summaries will explain patterns and results using the same information.

47. *The system shall provide operational reports covering available assets, outstanding requests, reservations, completed transfers, and maintenance activities, with filtering by department, category, and reporting period.*
48. *The system shall report reuse and repair activity and support estimates of avoided purchases, cost savings, waste diversion, asset-life extension, and carbon-emission reduction where the necessary data is available.*
49. *The system shall retain the inputs, calculation methods, units, factors, sources, and assumptions used for sustainability and financial estimates. Missing inputs shall be identified, and unsupported estimates shall not be presented as confirmed results.*
50. *The system shall distinguish completed activities from pending or cancelled actions. It shall prevent duplicate benefit claims for the same avoided purchase or disposal while retaining the full record of actual transfers and other events.*
51. *The system shall generate AI-assisted summaries of asset utilization, reuse activity, maintenance trends, departmental demand, estimated savings, and sustainability outcomes.*
52. *Numerical values in AI-generated reports shall agree with the underlying calculations. Generated reports shall identify their reporting period, data basis, and service or model version, and distinguish recorded outcomes from estimates.*

4.11 Notifications

Priority: Medium

Notifications will keep participants informed without requiring them to repeatedly check separate records. When a relevant event occurs, the system alerts the appropriate users and provides access to the information or action they need.

53. *The system shall provide in-application notifications about new matches, pending approvals, request and reservation status changes, transfer arrangements, ownership changes, and maintenance updates.*

- 54. *Each notification shall identify the relevant event, asset or request, and any required action, with direct access to the corresponding record or permitted workflow step.*
- 55. *Notification visibility shall respect the recipient's access permissions. Failure to deliver a notification shall not delete or alter the underlying request, approval, transfer, or maintenance record.*

5. Other Nonfunctional Requirements

These initial quality requirements will be reviewed during feasibility analysis and refined alongside the detailed requirements.

5.1 Performance Requirements

- 56. *Ordinary filtered searches shall complete within two seconds for at least 95% of measured requests using the declared test dataset and five simultaneous user sessions. The dataset and test environment shall be recorded. This provisional interaction budget is intended to limit waiting during repeated searches and will be checked during feasibility trials.*

5.2 Safety Requirements

- 57. *Tests involving assets recorded as unfit or incompatible with mandatory requirements shall confirm that ordinary-use reservation or release is prevented as applicable. AI assistance shall not bypass these checks or authorize physical actions.*

5.3 Security Requirements

- 58. *Authorization shall be enforced for every protected request, including direct access attempts and assistant interactions. Permission tests shall cover both allowed and denied actions for the defined roles.*
- 59. *Passwords shall not be stored in plaintext, and network access shall use encrypted connections. Credentials and sensitive configuration shall be excluded from the shared source repository.*

5.4 Software Quality Attributes

- 60. **Usability:** *Core publication, request, approval, and receipt tasks shall be completable using on-screen guidance without developer intervention. Forms shall support keyboard operation, identify errors clearly, and preserve entered information when validation fails.*
- 61. **Reliability:** *Confirmed records shall survive an application restart. Failed operations shall not leave partially applied reservations or transfers. Conventional workflows shall remain usable when AI services are unavailable, and longer AI operations shall display a processing status.*
- 62. **Maintainability:** *Setup instructions, configuration, and component interfaces shall be documented so another team member can run and maintain the prototype. AI integrations shall remain separable from the rules controlling authorization and asset transactions.*
- 63. **Resource efficiency:** *Conventional operations shall not invoke AI merely to enforce fixed business rules. AI-assisted operations shall record call counts, processing time, and estimated service cost where pricing information is available.*

5.5 Business Rules

Competing eligible requests will normally be considered in submission order; exceptions based on urgency, suitability, or timing require a recorded reason. Related approvals may be combined only within the representative's authority. Lack of demand or elapsed storage time will not automatically authorize disposal. Detailed delegation, retention, and exception policies will be refined in later phases.

6. Measurable Success Criteria

The prototype will be evaluated using documented test scenarios and labelled sample data. Expected outcomes, datasets, and test conditions will be recorded before evaluation.

Functional performance: All agreed critical workflow tests must produce the expected approvals, quantities, custody records, and final status. Tests will include competing reservations, cancellation, disputed receipt, and AI-service failure. Search performance will be checked against the provisional target in Section 5.

User acceptance: Reviewers who did not implement a task should be able to complete publication, requesting, approval, and receipt using the available guidance. Completion, errors, assistance required, time taken, and feedback will be recorded. Assisted data entry will be compared with manual entry to check whether it reduces effort without increasing errors.

AI usefulness: Conventional search and AI matching will be compared on the same requests, including synonymous descriptions, incompatible specifications, missing information, insufficient quantity, and no-match cases. Evaluation will count requests with a suitable resource among the first three results. The initial goal is improved retrieval on synonym cases without presenting confirmed incompatibilities as suitable. Classification suggestions and explanations will also be checked against the supplied information, recording correctness and user corrections.

Reuse, repair, and financial outcomes: Evaluation will include at least one completed internal-reuse case and one repair that restores an asset to service. Reported activity must agree with the underlying records. An avoided purchase will be counted only where a stated scenario establishes that reuse replaces a purchase; estimated net savings will deduct relevant transfer and repair costs.

Sustainability outcomes: Waste diversion, estimated asset-life extension, and avoided emissions will be reported using stated units, comparison assumptions, and factors. Calculations must agree with independently checked examples, avoid duplicate benefit claims, and identify missing inputs.

Sample-data evaluation demonstrates workflow and calculation performance, not actual university-wide savings or environmental impact. Targets and methods will be refined as feasibility findings and operational information become available.

7. Initial Project Plan

7.1 Development Methodology

The team will use iterative development within the course's phase milestones. Work will be divided into small tasks, reviewed regularly, and integrated progressively. Test cases will develop alongside requirements and implementation rather than being left until the final testing phase. Each milestone will produce a reviewed, versioned deliverable.

7.2 Timeline, Milestones, and Dependencies

The following planning dates follow the Phase 1 slides. Later dates remain provisional where the Word brief gives a different schedule.

Lab week, 2026	Milestone and deliverable
14 September	Phase 1: initial plan and requirement-gathering document
21 September	Phase 2: feasibility assessment
5 October	Phase 3: detailed requirements document
26 October	Phases 4–5: architecture and detailed design
16 November	Phase 6: working draft with major features
23 November	Phases 7–8: testing documentation, final project, presentation, and demonstration

Requirements and permissions will guide the data model and interfaces. Reliable asset and request records are prerequisites for matching, while completed activity records and agreed calculation inputs are needed for impact reporting. AI-service trials will begin during feasibility to identify integration and cost issues early.

7.3 Team Structure, Task Assignments, and Resources

The following allocation is an initial proposal for team review. Responsibilities are not fixed or exclusive and will be reviewed at each milestone according to members' skills, availability, and workload:

- **Abi Ali Murtaza Ghadiyali:** frontend/backend and AI integration, deployment planning
- **Ahmed Ghandour:** user interfaces and usability.
- **Abdulrahman Junaid:** data management, backend workflows, and access control.
- **Muhammad Awais Afzal Mirza:** AI features, evaluation, and reporting support.

All members will contribute to testing, documentation, and peer review. These workstreams establish initial ownership without preventing collaboration across features. The initial plan assumes access to team computers and a shared sample dataset. Programming tools, hosting, AI services, individual availability, and any service budget will be assessed during Phase 2. Paid services will not be assumed available before their cost and suitability are reviewed.

One GitHub repository holds phase documents, source code, tests, and project history. Members will use their own accounts and make regular, meaningful commits with clear messages. Changes will receive peer review, and submitted phase versions will be retained as later work develops.

7.4 Risks and Responses

Broad scope and limited time: Develop the required capabilities in dependency order and review progress against the brief at each milestone. Optional refinements will follow required functionality.

AI quality, cost, and data limitations: Trial candidate approaches early, use labelled examples, record limitations, and retain conventional workflows when assistance is unavailable.

Integration and record inconsistency: Agree interfaces during design, integrate regularly, and test permissions, competing reservations, transfers, and calculations across components.

Unconfirmed availability and responsibilities: Confirm the proposed allocation during team review and redistribute work when dependencies or workload create bottlenecks.

Feasibility will also clarify permitted tools, data sources, operational assumptions, sustainability factors, and the expected depth of future-demand prediction. Detailed approval exceptions and retention policies will be refined during requirements analysis.

Sources: This report draws on the supplied Fall 2026 project brief, Phase 1 lab slides, SRS template, and Chapter 1 lecture material.